Dynamic Planet C - Dynamic Planet - Highlands Virtual Invitational - 10-08-2022

There is an Image Key for this event. This is the link: Image Key (https://drive.google.com/file/d/1mJeUnwcBGia8HyZj9py4rBTcCIM0iTWA/view?usp=sharing)

1. (1.00 pts) Refer to image 1a, what cycle is shown in the image above?

Expected Answer: Hydrologic/Water Cycle

2. (18.00 pts) From A to R label the parts with the correct term/process: (1 pt per correct answer)

Expected Answer: A: Water storage in snow and ice B: Precipitation C: Snowmelt Runoff D: Infiltration E: Groundwater Flow/ Groundwater Discharge (either one works) F: Groundwater storage G: Evaporation from Ocean H: Evaporation I: Condensation J: Water Storage in the Atmosphere K: Evapotranspiration L: Sublimation M: Depression N: Streamflow O: Spring P: Freshwater Q: Surface Runoff R: Plant uptake

3. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3a?

Expected Answer: Dendritic

4. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3b?

Expected Answer: Parallel

5. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3c?

Expected Answer: Trellis

6. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3d?

Expected Answer: Radial

7. (1.00 pts) Alongside your answer to question 3d. What geologic feature is this drainage/stream pattern normally found at?

Expected Answer: Volcanoes or cone-shaped mountains

8. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3e?

Expected Answer: Rectangular

9. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3f?

Expected Answer: Centripetal

10. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3g?

Expected Answer: Deranged or contorted

11. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3h?

Expected Answer: Braided

12. (1.00 pts) Refer to the Images on the Image Key. What stream channel is Image 3i?

Expected Answer: Anastomosed River

13. (12.00 pts) Referring to Image 4a. Label the features of the meandering channel and define it (2 pts per answer, label and definition):

Expected Answer: A: Meander: a winding curve or bend of a river B: Oxbow Lake: A separate body of water from the lake C: Cutoff: occurs when the stream erodes through the neck causing the river to be back to a straight course D: Neck: the point of land between two edges of a meander E: Natural Levee: embankments formed naturally after a river floods and recedes F: Alluvium: A deposit of clay, silt, sand, and gravel left by flowing streams in a river valley or delta, typically producing fertile soil.

14. (4.00 pts) Describe the process of the formation of a meandering stream.

Expected Answer: Starts off as a narrow v-shaped valley. Lateral erosion takes place and the floodplain becomes more developed. The water turns and bends to avoid obstacles. A speed slows down the sediment drops down on one side and then picks up more on the other side. Meanders become more sinuous with time. The cut bank erodes; the point bar accretes. Meander curves become more pronounced and elongated. Erosion and deposition of meanders over long time scales create a broad valley bottom

15. (1.00 pts) Define **Porosity**

Expected Answer: Porosity: The total volume of open space

16. (1.00 pts) Define **Permeability**

Expected Answer: Permeability: the ease of water flow due to pore interconnectedness

17. (1.00 pts) Referring back to the definition you answered for **permeability,** what increases the permeability of a waterway?

Expected Answer: Large and straight flow

18. (1.00 pts) Define **Aquifers**

Expected Answer: Aquifers: Porous and permeable rocks or sediment that freely transmit groundwater

19. (1.00 pts) Do aquifers have high or low porosity and permeability?

Expected Answer: High

20. (1.00 pts) Define **Aquitards**

Expected Answer: Aquitards: Impermeable layers that hinder or prevent water movement

21. (1.00 pts) Do aquitards have low or high porosity and permeability?

Expected Answer: low

22. (1.00 pts) Define **Sinuosity**

Expected Answer: Sinuosity: The ability to curve or bend easily and flexibly

23. (2.00 pts) What are the differences between Permanent and Ephemeral Streams?

Expected Answer: Permanent (Gaining) Streams Water flows all year At or below the water table Humid or temperature Sufficient rainfall Lower evaporation Seasonal discharge variation Ephemeral (Losing) Streams Do not flow all year Above the water table Arid or semi-arid climates Low rainfall High evaporation Flows mostly during rare flash floods

24. (2.00 pts) What is stream discharge and what is the formula for it?

Expected Answer: The amount of water flowing in a stream channel D = AC* Va D = discharge Ac = cross sectional area Va = average velocity

25. (3.00 pts)
Rivers are one of the reasons for shaping the Earth's surface. It does this through Erosion, sediment transport and deposition. What is the dissolved load, suspended load, and bed load?

Expected Answer: Dissolved load: ions from mineral weathering Suspended load: fine particles (silt and clay) in the flow Bed load: coarser particles that move along the stream bed

26. (2.00 pts) What is stream capacity and what is it controlled by?

Expected Answer: Capacity: The amount of sediment a stream can transport Controlled by discharge (Q)

27. (1.00 pts) What is stream competence?

Expected Answer: The maximum size of sediment that a stream can transport that is controlled by velocity.

28. (2.00 pts) Referring to Image 12a on the image sheet, what is the formula for discharge and how do you find it on a hydrograph?

Expected Answer: Discharge (Q) = Volume / Time Look at the area under the curve of a flood hydrograph. The total volume of water present.

29. (1.00 pts) What is meant by the term ‘lag time’?

Expected Answer: The time it takes for the water to get to the stream. The difference in time between the height of the storm and the highest flow of the river

30. (1.00 pts) What effect does urbanization have on a hydrograph?

Expected Answer: Urbanization forms impermeable surfaces, creates a steep rising limb and shortens the lag time

31. (1.00 pts) What does a ‘rising limb’ on a hydrograph indicate?

Expected Answer: How quickly the level of water in a river rises after a storm

32. (1.00 pts) What is a recurrence interval?

Expected Answer: A return period, also known as a recurrence interval or repeat interval, is an average time or an estimated average time between events such as earthquakes, floods, landslides, or river discharge flows to occur.

33. (1.00 pts) If a 100 year flood occurs on the Mississippi river in 2010, what is the probability that a flood of the same magnitude will occur in 2022?

Expected Answer: 0.114

34. (2.00 pts) Refer to Image 18a, Which line represents an Ephemeral stream, which one is a permanent stream?

Expected Answer: Red; Ephemeral Blue: Permanent

35. (1.00 pts) What is percolation?

Expected Answer: flux of water within the soil

36. (1.00 pts) Larger grains have _____ infiltration rate and _____ percolation rate?

- A) Lower/ lower
- B) higher/lower
- C) lower/higher
- D) higher/higher

37. (3.00 pts) Referring to Image 21a, what type of landform is this, and how is it formed?

Expected Answer: Deltas form when a stream enters the standing water. The current slows and loses competence; sediments drop out. The stream divides into a fan of small distributaries. Shape due to the interplay of flow, waves, and tides.

38. (1.00 pts) Define **cave**

Expected Answer: Caves: are open cavities in rock- most are formed by the dissolution of limestone bedrock

39. (1.00 pts) Define **Karst Landscape**

Expected Answer: Karst Landscape: are formed in areas of abundant limestone bedrock and caves

40. (1.00 pts) Define **speleothems**

Expected Answer: Speleothems: a structure formed in a cave by the deposition of minerals from water, e.g. a stalactite or stalagmite.

41. (1.00 pts) Refer to Image23a, what does this USGS topographic symbol represent?

Expected Answer: Perennial stream

42. (1.00 pts) Refer to Image 24a, what does this USGS topographic symbol represent?

- A) Perennial stream
- B) Intermittent river
- C) Perennial river
- D) Intermittent stream

43. (1.00 pts) Refer to Image 25a, what type of feature is represented by this USGS topographic symbol?

- A) Bridge
- B) Highway
- C) Trail
- D) Levee

44. (1.00 pts) Refer to Image 26a, what freshwater feature has the following symbol on a USGS topographic map?

- A) Narrow wash
- B) Marsh or swamp
- C) Spring or seep
- D) B and c

45. (1.00 pts) Refer to Image 27a, what landform is this?

Expected Answer: Alluvial Fan

46. (2.00 pts) What is the difference between cover subsidence and and cover collapse?

Expected Answer: Cover subsidence Downward movement of earth to fill a void Can be a natural process or human-induced Human-induced: oil/natural gas extraction, mining, groundwater-related Natural: dissolution of limestone Generally, years in the making but maybe a sudden collapse Cover collapse Where impermeable overburden contains clay and sits atop limestone, soil cavities grow large, become unstable, and collapse suddenly into steep-sided sinkholes

47. (3.00 pts) What is the process for a cover collapse sinkhole?

Expected Answer: Process: Sediments spall into a cavity. As spalling continues, the cohesive covering sediments form a structural arch. The cavity migrates upward by progressive roof collapse. The cavity eventually breaches the ground surface, creating sudden and dramatic sinkholes.

48. (1.00 pts) How are drainage basins formed?

Expected Answer: A drainage basin is formed by the action of water as it forms streams and rivers that flow downhill

49. (1.00 pts) What percentage of glaciers and ice caps store all water on Earth?

- (●) A) 2%
- ( ) B) 3%
- ( ) C) 5%
- ( ) D) 10%

50. (1.00 pts) Which of the following sediment types would have the greatest porosity to permeability ratio?

- (●) A) Clay
- ( ) B) Silt
- ( ) C) Sand
- ( ) D) Gravel
- ( ) E) Cobble

51. (21.00 pts) List all 21 types of lakes

Expected Answer: Cirque/tarn lakes Glacial lakes Rock hollow lakes Volcanic lakes Meteorite lakes Solution lakes Fluvial lakes Oxbow lakes Landslide lakes Tectonic lakes Aeolian lakes Freshwater lakes organic/animal-made lakes Barrier lakes Saline lakes Rift valley lakes Karst lakes Wind deflation lakes Human made/ artificial lakes Temporary lakes Permanent lakes

52. (1.00 pts) What does Darcy's Law state?

Expected Answer: Darcy's law states that the negative of permeability/dynamic viscosity of the fluid multiplied by the pressure drop is equal to the instantaneous flow rate of a fluid.

53. (1.00 pts) What is the main difference between a braided stream channel and an anastomose channel?

Expected Answer: Anastomose channels are basically braided stream channels, but they don't divide or reconnect as easily as braided stream channels.

54. (4.00 pts) Define potential evapotranspiration and describe at least three parameters that it depends on.

Expected Answer: Potential evapotranspiration is the maximum amount of water vapor that could be evaporated or transpired given the atmospheric conditions (wind speed, air temperature, insolation, relative humidity)

55. (1.00 pts) Which of the following are reasons for removing dams from rivers (mark all that are correct)?

(Mark **ALL** correct answers)

- [x] A) Allowing fish migration
- [ ] B) Reduce flooding
- [x] C) Filling of the reservoir behind the dam with sediment
- [ ] D) Reducing the effects of upstream land impacts on stream water quality

56. (1.00 pts) True or False: Inflow of groundwater contaminated with agricultural chemicals to an estuary is an example of nonpoint pollution.

- (●) True
- ( ) False

57. (1.00 pts) Define **Pingo**

Expected Answer: Pingo: :A hill with a core of ice, found in permafrost environments.

58. (1.00 pts) Define **Permafrost**:

Expected Answer: Permafrost: Ground that is permanently frozen

59. (1.00 pts) What two processes are most likely to lead to a rise in the water table

- A) Erosion and evaporation
- B) Infiltration and evaporation
- C) Erosion and precipitation
- D) Infiltration and precipitation

60. (1.00 pts) On a topographic map, the point of the “V” shapes in contour lines around rivers and streams point_____.

- A) Downstream
- B) Toward the center
- C) Toward the banks
- D) Upstream

61. (1.00 pts) Name 4 types of waterfalls

Expected Answer: Punchbowl, plunge, multi-step, horsetail, frozen, fan, chutes, cataracts, cascades, block

62. (2.00 pts) Define **Epilimnion** and **Hypolimnion**

Expected Answer: Epilimnion: The upper layer of water in a stratified lake Hypolimnion: the lower layer of water in a stratified lake, typically cooler than the water above and relatively stagnant.

63. (1.00 pts) What separates the epilimnion from the hypolimnion?

Expected Answer: Either thermocline or metalimnion

64. (1.00 pts) Which lake zone is sometimes referred to as the “lake’s digestive system,” as bacteria decompose organic matter in this area?

Expected Answer: Benthic Zone

65. (1.00 pts) What term describes the average time that it takes to completely renew the water volume of a lake?

Expected Answer: Flushing time

66. (1.00 pts) Approximately how much of the extracted groundwater on Earth is used for agricultural purposes?

- A) 15%
- B) 25%
- C) 40%
- D) 70%

67. (1.00 pts) Which term refers to a dry lake bed?

- A) Playa
- B) Arroyo
- C) Thalweg
- D) Eddy

68. (1.00 pts) At UCI which term from above is a dorm name at Mesa Court? And what language does it come from? (TB)

Expected Answer: Playa, Spanish

69. (1.00 pts) A clast of 1.2 mm is classified as a?

A) Silt
B) Clay
C) Sand
D) Gravel

70. (1.00 pts) A river is flowing at a velocity of 12 cm/s. What kind of sediments can be transported?

A) Clay
B) Clay and silt
C) Clay, silt, gravel
D) Clay, silt, sand

71. (1.00 pts) Which of the following can cause the formation of knickpoints?

A) Tectonic uplifting
B) Decreased ocean water influx
C) Isostatic rebound
D) All of the above

72. (1.00 pts) Which of the following provide evidence for river capture?

A) Cols
B) Water grapes
C) Under-fit streams
D) All of the above
E) None of the above

73. (1.00 pts) Which of the following processes does NOT follow permafrost melting?

A) Mountain slope stabilization
B) Release of natural metal deposits

C) Swamp and lake formation

D) Emission of methane and carbon dioxide

74. (1.00 pts) Define **Groundwater**

Expected Answer: Groundwater is water that occurs below the ground surface in zones of porous materials where all the cracks (fractures) and pore spaces are filled with water and that water is at a pressure higher than the pressure of the atmosphere.

75. (1.00 pts) How much of the hydrosphere is groundwater?

Expected Answer: ~1.7%

76. (1.00 pts) Which factor influences the quality of drinking water the most?

A) pH

B) Dissolved Oxygen

C) TDS

D) Temperature

77. (1.00 pts) Suppose a certain stream's load is mainly sand and gravel. Would you expect this stream to be a:

A) Meandering channel

B) Bedrock channel

C) Braided channel

D) Alluvial channel

78. (1.00 pts) Suppose a certain channel has potholes. Is this channel more likely a:

A) Alluvial channel

B) Braided channel

C) Bedrock channel

D) Meandering channel

79. (1.00 pts) A stream that doesn't meander at all has a sinuosity of____?

Expected Answer: 1

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